



Xilinx' Stacked Silicon Interconnect technology

cent less power than traditional ICs.

7. 3D stacking permits heterogeneity, since the dice arranged one above the other need not be similar. This allows a designer to optimise components according to their function.

Are today's '3D' chips really 3D?

There is a difference between a 3D package and a 3D IC.

In a 3D package, separate chips are stacked in a single package. They may be stacked one above the other or side-by-side. This arrangement is often called a chip stack, system-in-package (SiP) or multi-chip module (MCM). Here the chips are merely stacked and connected using interconnects but not integrated into a single chip.

In contrast, a 3D IC is a single chip. Multiple dice are stacked, connected using through-silicon vias (TSVs), and multiple groups of dice again connected together using a silicon interposer—and all of this is mounted on a single SiP.

3D chips are in development stage, and any chip touted as 3D today is essentially a 2.5D chip or a 3D package as most of the commercially-available or practicable designs today are MCMs or package-on-package. Pure 3D ICs (stacking in package) are still a point of research in most companies, and the big foundries have just figured out the processes and interposer technology

for the same.

Although Intel has showcased quite a few 3D ICs in the past, such as the 3D Pentium CPU in 2004 and the Teraflops Research Chip in 2007, the area continues to haunt the silicon giant because of the difficulty in heat dissipation and the high cost of manufacturing involved. This year too, 3D-IC-based multi-core designs using GlobalFoundries' 130nm process and Tezzaron's FaStack technology were demonstrated at a leading conference, as also some designs by teams from the Georgia Institute of Technology and the University of Michigan. However, it is unclear when the technology might actually reach production.

Techniques for stacking the dice in a 3D chip

There are several ways to stack chips. Giles Peckham, EMEA director-Marketing, Xilinx, explains the most common ones:

1. Connecting two stacked chips to a flip chip at the bottom of the stack. Examples are SoCs where dynamic random-access memory (DRAM) is placed on top of the central processing unit (CPU).

2. Using TSV, where the dice are placed on top of each other and connected through vertical copper channels. An example of TSV usage is the super-density DRAMs used in camera

CMOS sensors.

3. Making use of a silicon transposer, which connects two or more chips together—much like the transposers used in AMD graphics cards.

Xilinx adopts a technology called stacked silicon interconnect (SSI). In this architecture, multiple dice are placed side-by-side on top of a passive silicon interposer, which facilitates interconnect/communication between the many dice. Such a design, where the chips are placed side by side, is called a multi-chip module but should not be confused with a traditional MCM in which the connections between the chips are made via powerful I/O buffers and bond wires. In SSI technology, the connections are virtually seamless, the interposer being an extension of the on-chip interconnect with microbumps to join them. With more than 10,000 interconnects that can be programmed among the dice, and programmability of each die and the I/O, the company claims its ICs to be one of the first commercially-available, completely-programmable 3D ICs.

Texas Instruments has an alternative technology, called PowerStack, for 3D packaging. It involves stacking TI's NexFET power MOSFETs on a grounded lead frame and connecting them using copper clips. This combination of stacking and clip bonding results in a more integrated quad flat no-lead (QFN) solution that delivers a smaller size, better thermal performance, higher switching speed and higher current capabilities over more traditional solutions that utilise MOSFETs placed side-by-side.

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